**REVISION WORKSHEET**

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

# AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

## Impulse and Momentum Practice Questions

**Grade 11 Advanced | Module 9 Lesson 1 | Impulse and Momentum**

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $p = mv$; $J = F_{avg}\Delta t = \Delta p$; $L = I\omega$; $\tau_{net}\Delta t = \Delta L$. Use signed velocities and consistent directions.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 30–37.

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**P01. Linear momentum is:**

A. Mass divided by speed B. Mass multiplied by velocity
C. Force divided by time D. Energy multiplied by distance

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**P02. The area under a force–time graph represents:**

A. Velocity B. Position
C. Impulse D. Power

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**P03. For a fixed change in momentum, increasing stopping time reduces:**

A. Impulse magnitude B. Mass necessarily
C. Change in velocity necessarily D. Average stopping force

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**P04. Angular momentum about a fixed axis is:**

A. $I\omega$ B. $I/\omega$
C. $\tau/\omega$ D. $mr$

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**P05. The SI unit N s is equivalent to:**

A. kg $m^2/s^2$ B. kg m/s
C. kg/s D. $m/s^2$

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**P06. A 1 kg object moves at +4.0 m/s. Calculate its linear momentum.**

A. 8 kg m/s B. 2 kg m/s
C. 4 kg m/s D. -4 kg m/s

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**P07. A constant force +40 N acts for 0.050 s. Find its impulse.**

A. 4 N s B. 1 N s
C. -2 N s D. 2 N s

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**P08. A 0.4 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

A. 80 N B. 160 N
C. 40 N D. -80 N

**P09. A rotor has I = 0.4 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

A. 4.8 kg m²/s  B. 2.4 kg m²/s
C. 1.2 kg m²/s  D. -2.4 kg m²/s

**P10. A constant net torque +6 N m acts for 0.20 s. Find the change in angular momentum.**

A. 2.4 kg m²/s  B. 0.6 kg m²/s
C. 1.2 kg m²/s  D. -1.2 kg m²/s

**P11. Define impulse.**

**P12. State the impulse–momentum theorem.**

**P13. Define angular momentum for rotation about a fixed axis.**

**P14. Why can airbags reduce injury in a crash?**

**P15. State the angular impulse–angular momentum theorem.**

**P16. A 1.5 kg object moves at +4.0 m/s. Calculate its linear momentum.**

**P17. A 2 kg object moves at +4.0 m/s. Calculate its linear momentum.**

**P18. A constant force +60 N acts for 0.050 s. Find its impulse.**

**P19. A constant force +80 N acts for 0.050 s. Find its impulse.**

**P20. A 0.6 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

**P21. A 0.8 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

**P22. A rotor has I = 0.6 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

**P23. A rotor has I = 0.8 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

**P24. A constant net torque +9 N m acts for 0.20 s. Find the change in angular momentum.**

**P25. A constant net torque +12 N m acts for 0.20 s. Find the change in angular momentum.**

**P26. A force–time graph is triangular with base 0.020 s and peak force 300 N. Calculate impulse.**

**P27. A ball moving toward a bat rebounds in the opposite direction. Why must signs be used in calculating Δp?**

**P28. Compare catching a ball with rigid hands and with hands moving backward. Which usually reduces force and why?**

**P29. An ice skater pulls in their arms with negligible external torque. Explain what happens to angular speed.**

**P30. A large force acts for a very short time. Is its impulse necessarily greater than a small force over a long time?**